

TERMUX INSTALL AND SETUP - Complete Guide for Mobile Linux Environment

◇ Introduction

Termux Android ke liye ek powerful terminal emulator aur Linux environment hai jo bina root access ke Linux commands aur packages chalane ki facility deta hai. Iski madad se aap programming languages, development tools, SSH, Git aur alag-alag Linux distributions ko Android device par use kar sakte hain.

◇ **TERMUX KYA HAI?**

Termux ek Android application hai jo Linux command-line environment provide karti hai. Iske through aap Linux packages install kar sakte hain, shell commands chala sakte hain aur development ya learning ke liye ek portable Linux environment use kar sakte hain.

Termux mein Python, Node.js, Git, SSH aur bahut se open-source tools install kiye ja sakte hain. Agar zaroorat ho to supported Linux distributions ko **proot-distro** ke through bhi chalaya ja sakta hai.

◇ **TERMUX INSTALLATION**

Recommended Download Source

Termux ko **F-Droid** se install karna recommended hai kyunki wahan latest stable version available hota hai. Google Play Store wala version kaafi purana hai aur actively maintain nahi kiya jata.

Official sources:

F-Droid (Recommended)

<https://f-droid.org/en/packages/com.termux/>

GitHub Releases

<https://github.com/termux/termux-app/releases>

Google Play Store version ko use karna generally recommend nahi kiya jata kyunki usme latest updates aur features available nahi hote.

◇ **SYSTEM REQUIREMENTS**

Android 7.0 ya usse newer version recommended hai. ARM, ARM64 (AArch64), i686 aur x86_64 architectures supported hain.

Kam se kam 300 MB free storage aur lagbhag 1 GB RAM recommended hai.

Agar APK manually install karte hain to Google Play Protect warning aa sakti hai. Agar aap trusted official source se download kar rahe hain to installation continue kar sakte hain.

◇ INITIAL SETUP

Step 1 – Packages Update Karein

Termux install karne ke baad sabse pehle packages update karein.

```
pkg update && pkg upgrade -y
```

Ye command package list update karti hai aur installed packages ko latest version par upgrade karti hai.

Step 2 – Storage Permission Grant Karein

Android storage access enable karne ke liye:

```
termux-setup-storage
```

Permission popup aane par **Allow** select karein.

Iske baad \$HOME/storage directory create ho jati hai jahan se shared storage access ki ja sakti hai.

Storage verify karne ke liye:

```
ls ~/storage
```

Step 3 – Essential Packages Install Karein

Wget install karein:

```
pkg install wget -y
```

Git install karein:

```
pkg install git -y
```

Curl install karein:

```
pkg install curl -y
```

Python install karein:

```
pkg install python -y
```

Proot-Distro install karein:

```
pkg install proot-distro -y
```

OpenSSH install karein:

```
pkg install openssh -y
```

Step 4 – Additional Repositories

Agar additional repositories ki zaroorat ho to:

```
pkg install tur-repo -y
```

Aur X11 repository install karne ke liye:

```
pkg install x11-repo -y
```

Step 5 – Repository Mirror Change Karna

Agar package download speed slow ho to repository mirror change karein.

```
termux-change-repo
```

◇ **LINUX DISTRIBUTION INSTALL KARNA**

Termux ke andar **proot-distro** ke through supported Linux distributions install ki ja sakti hain.

Proot-Distro install karein:

```
pkg install proot-distro -y
```

Kali Linux install karein:

```
proot-distro install kali
```

Kali Linux login karein:

```
proot-distro login kali
```

Ubuntu install karne ke liye:

```
proot-distro install ubuntu
```

Fedora install karne ke liye:

```
proot-distro install fedora
```

◇ **BASIC LINUX COMMANDS**

Packages update aur upgrade:

```
pkg update && pkg upgrade -y
```

Storage setup:

```
termux-setup-storage
```

Repository change:

```
termux-change-repo
```

Package install:

```
pkg install <package_name>
```

Package search:

```
pkg search <package_name>
```

Files list karne ke liye:

```
ls
```

Directory change karne ke liye:

```
cd <directory>
```

Current directory dekhne ke liye:

```
pwd
```

Current user check karne ke liye:

```
whoami
```

System information dekhne ke liye:

```
uname -a
```

◇ **COMMON ISSUES AND SOLUTIONS**

Storage Permission Problem

Agar storage access kaam nahi kar rahi ho to dobara ye command chalayein:

```
termux-setup-storage
```

Aur Android permission settings mein jaakar storage permission verify karein.

Repository Slow Ho

Agar packages download hone mein bahut time lag raha ho to:

```
termux-change-repo
```

Aur nearest mirror select karein.

Package Install Nahi Ho Raha

Sabse pehle packages update karein:

```
pkg update && pkg upgrade -y
```

Uske baad package dobara install karne ki koshish karein.

Google Play Protect Warning

Agar official F-Droid ya GitHub APK install karte waqt Play Protect warning aaye to source verify karke installation continue ki ja sakti hai.

◇ **RECOMMENDED PACKAGES**

Development aur learning environment ke liye ye packages useful hote hain:

```
pkg install git curl wget python openssh nano vim  
zip unzip -y
```

Ye packages file management, programming, networking aur command-line kaam ke liye commonly use kiye jate hain.

⚠ IMPORTANT NOTES

Termux ko hamesha official sources (F-Droid ya GitHub Releases) se hi download karein.

Regularly packages update karte rahein taaki latest security patches aur bug fixes milte rahein.

Linux commands aur tools sirf un systems aur devices par use karein jinke liye aapke paas authorization ho.

⚠ EDUCATIONAL NOTE

Ye notes sirf educational aur learning purposes ke liye hain. Kisi bhi system ya device par bina permission ke access, testing ya modification karna applicable cyber laws ke under punishable offense ho sakta hai.

MADE BY:
SYED GHOUS
